

WASP-11b

R-1451

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_141124 · created 2026-08-01T09:33:57+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456985.7312 +/- 0.00089 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1423 +/- 0.0052
Transit depth (Rp/Rs)^2	2.03 +/- 0.15 [%]
Orbital Inclination (inc)	89.97 +/- 0.34
Airmass coefficient 1 (a1)	0.9934 +/- 0.0062
Airmass coefficient 2 (a2)	0.005 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.62 %
Best Comparison Star	#3 - [550, 357]
Optimal Aperture	3.81
Optimal Annulus	10.98
Transit Duration (day)	0.10813 +/- 0.00058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1423 ± 0.0052 vs 0.131 (deviation 2.17σ)
Duration fit vs published	0.10813 d vs 0.1075 d (deviation 1.09σ)
Mid-time O–C	1.8 min
Depth SNR	27.4
Residual scatter	0.62 %

Light curve

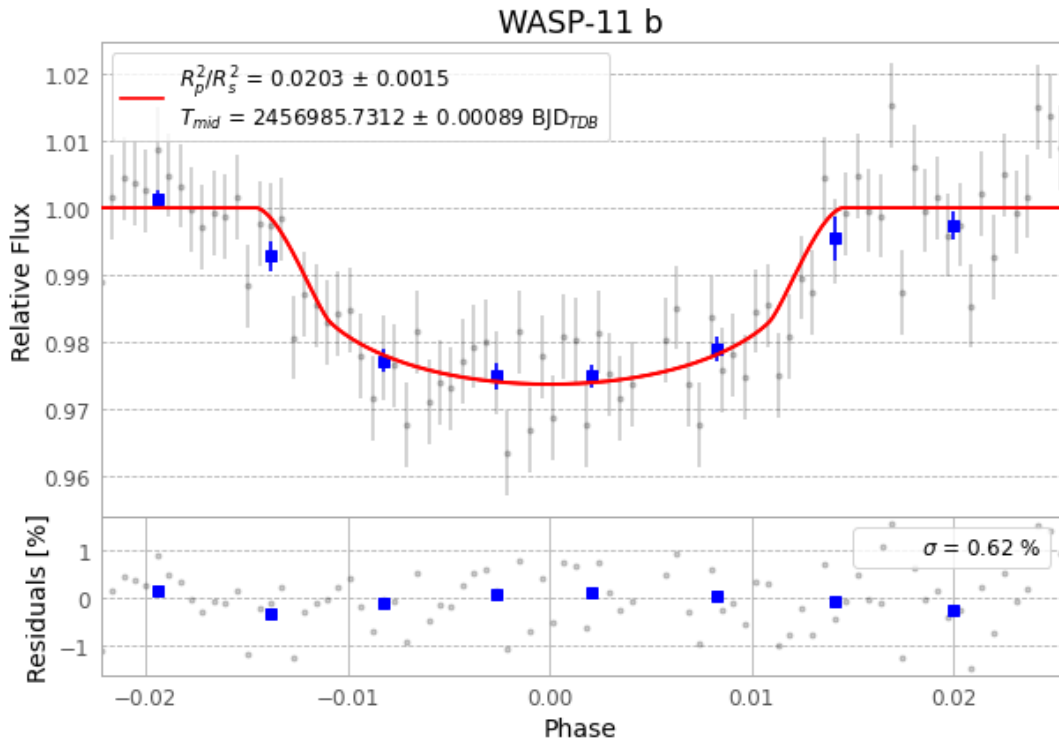


Figure 1 — FinalLightCurve_WASP-11 b_2014-11-23.png (SHA-256 0b8f4da56b3513e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [490, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2f2ea0a1daca4612...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), HATP-10141124032412.FITS → HATP-10141124073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-141124.log (094f9c140957...), FinalLightCurve_WASP-11 b_2014-11-23.png (0b8f4da56b35...), FinalLightCurve_WASP-11 b_2014-11-23.csv (ba5b7eec7a28...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 09:33Z.